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Priyotosh Bandyopadhyay

Curriculum Vitae

Present Position

- October, 2021- Associate Professor, Dept. of Physics at Indian Institute of Technology Hyderabad(IITH)
2023-24 Visiting Professor, Korea Institute for Advanced Study, Seoul, South Korea

Previous Positions

- July, 2016-October, 2021 Assistant Professor Grade-I, Dept. of Physics at Indian Institute of Technology Hyderabad(IITH)
November, 2014- July, 2016 Istituto Nazionale di Fisica Nucleare (INFN) Fellow at University of Salento, Lecce, Italy
Dec, 2011-October, 2014 Postdoctoral Fellow at University of Helsinki, Helsinki, Finland
Dec, 2011-October, 2014 Adjunct Scientist at Helsinki Institute of Physics, Helsinki, Finland
Jan, 2010-Dec, 2011 Postdoctoral Fellow at Korea Institute for Advanced Study, Seoul, South Korea

Education

- 2004-2009 **Doctor of Philosophy**, *Harish-Chandra Research Institute (HRI)*, Allahabad (India), *Homi Bhabha National Institute(HBNI)*.
2002-2004 **Master of Science (Physics)**, *I.I.T(Mumbai)*, Powai, Mumbai(India), 1st class. CPI: 8.95 (in 10 point grade), rank- 2nd
2003 I was selected as VSRP student and I have done a project with Prof. Deepak Dhar at TIFR, Mumbai
1999-2002 **Bachelor of Science, Honours in Physics**, *R.K.M.Vidyamandir, University of Calcutta*, (India), 1st Class. Physics(H), 73.29% Total

Doctoral Thesis

Title	Some Studies on Higgs searches at the Large Hadron Collider in the scenarios beyond the Standard Model
Supervisor	Prof. Aresh Krishna Datta, Harish-Chandra Research Institute, Allahabad
Thesis Submitted	September, 2009
Degree Received	February, 2010

Area of Specialization

Electroweak Phase Transition and Gravitational wave, Neutrino Physics, Dark matter, Physics at the different colliders

Teaching Experience

At IITH

2025-2026, (15 crs)	Classical Mechanics, first year MSc physics students (3cr). Advanced Particle Physics(3cr), Advanced topics in High Energy Physics(3 crs), Beyond the Standard Model (3 cr), Supersymmetry (3cr) for MSc, EP3, EP4, Ph.D students.
2024-2025, (10 crs)	Classical Mechanics, first year MSc physics students (3cr). Topics Beyond the Standard Model: Ph.D, MSc and EP students (1cr) Computational Particle physics, For MSc, EP students and Ph.D students (3cr). B Tech Lab: For BTech across the department (3cr)
2023-2024, (6 crs)	Electrodynamics II, 3 crs: For MSc physics, medical physics and EP students. Advanced Particle physics, 3 crs: For MSc, EP students and Ph.D students.
2022-2023, (6 crs)	Electrodynamics II, 3 crs: For MSc and EP students Computational Particle physics, 3 crs: For MSc and EP students
2021-2022, (6 crs)	Electrodynamics II, 3 crs: For MSc and EP students Advanced Particle physics, 3 crs: For MSc and EP students
2020-2021, (6 crs)	Nuclear Physics (1cr): for M.Sc students and Engineering Physics students. Modern Physics (2cr): First year B.Tech. Computational Particle Physics/HEP tools(3cr and 2 crs): Ph.D/M.Sc and EP students B.Tech project evaluation for EP

Supervising two project and Four Ph.D students
Anomalies 2020 and international conference

- 2019-2020, (6 crs)
Quantum Physics (3 crs): for Ph.D students.
Relativity(1 cr): for Engineering Physics students.
Advanced Particle Physics (2 crs): MSc, Phd
B.Tech project evaluation for.
Supervising four project and three Ph.D students
- 13-18/01/2020, Invited lectures
Extended Higgs scenarios in the case of Non-SUY and SUSY, total 4 lectures given for the Ph.D students at Institute of Physics, Bhubaneswar
- 2018-19, (7 crs)
Quantum Physics (3 crs):: for Ph.D students.
Relativity (1 cr): for Engineering Physics students.
Tensors and Differential forms (1 cr): for Engineering Physics students.
Advanced Particle Physics (2 crs): MSc, Phd
Supervising three project and three Ph.D students
I supervised was actively involved in organising our second Physics Open Day.
- 2017-18, (7 crs)
Quantum Physics (3crs):: for Ph.D students.
Relativity (1 cr): for Engineering Physics students.
Tensors and Differential forms(1 cr): for Engineering Physics students.
Advanced Particle Physics (2 crs): for Masters and Engineering Physics students.
M.Sc project evaluation
Supervising four project and one Ph.D students
I supervised was actively involved in organising our First Physics Open Day.
- 2016-17, (9 crs)
Quantum Physics(3 crs):: for Ph.D students.
Electrodynamics(2 crs):: for Engineering Physics students.
Tensors and Differential forms(1 cr):: for Engineering Physics students.
B.Tech Lab(3 crs)
M.Sc project evaluation
Supervising four project and one Ph.D students

[During Postdoc](#)

- 2015: University of Helsinki
HepTools2: I gave a short course on computational particle physics as a guest lectures at University of Helsinki. The course was attended by Ph.d students and postdocs.

- 2014: University of Helsinki Heptools : During Summer (2013) I taught a course on numerical simulation in particle physics: CalcHep, PYHTIA and other packages. Spring 2014, a formal course 'Heptools' was given by me and students from University of Helsinki credited the course.
- 2013: University of Helsinki Autumn Semester 2013: Organising Particle physics seminar series.
- 2013: University of Helsinki Spring semester 2013, I am giving a lectures series and organising a seminar series on 'Special topics in particle physics: the era of LHC part II'
- 2012: University of Helsinki I taught the course 'Special topics in particle physics: the era of LHC' for the Ph.D students in Helsinki Institute of Physics and Helsinki University for the spring semester, 2012.
- 2010-11: KIAS Organiser of weekly journal club at KIAS, Seoul South Korea.
- During Ph.D I was TA for the Quantum Field Theory-II, by Ashoke Sen during my PhD at HRI

Supervising Ph.D Students At IITH

- 2016-2025 Currently I am supervising three Ph.D students. So far four Ph.D students already graduated.
- Aug,2017-April, 2021 **Dr. Saunak Dutta, graduated.** Saunak Dutta finished his Ph.D in 3 years and 9 months. He was working as assistant professor at Vijaybhoomi School of Science and Technology and now at Deloitte Tohmatsu Financial Advisory LLC
- Aug,2019-September, 2022 **Dr. Shilpa Jangid , graduated.** Shilpa Jangid submitted her thesis in September, 2022 and defended her thesis in February, 2023 and joined APCTP, Pohang as post-doctoral fellow. She finished her Ph.D thesis within 3.3 years. Currently she is at Shiv Nadar University, Noida, UP, India as a postdoc.
- Aug,2019- Aug, 2024 **Dr. Chandrima Sen, graduated.** She submitted her thesis in 22nd August, 2024 and defended her thesis on 26th September, 2024. Currently working as post doctoral researcher at Physical Research Laboratory, Ahmedabad.
- Aug,2020-September, 2025 **Dr. Snehashis Parashar, graduated:** He submitted his thesis on 8th of September, 2025 and defended his thesis on October, 27th, 2025. He is about to join Institute of Physics Bhubaneswar as a postdoc.
- Jan,2022- Pram M. Robin joined as Ph.D student in January, 2022. He has been working on Fixed point analysis of BSM models. He is about to finish in a year.
- July, 2025- Disha Ghosh joined as Ph.D student in July, 2025. She is working on the thermal field theory and first order phase transition in the extended scalar sectors. She is also pursuing the Ph.D course work.

August, 2025- Gaurav Joshi joined as Ph.D student in August, 2025. He is working on Anomalies, leptogenesis and baryogenesis in beyond Standard Model scenarios. He is also pursuing the Ph.D course work.

At INFN Lecce

2014-16 I co-supervised Antonio Costantini, student of Prof. Claudio Coriano, towards his Ph.D thesis during 2014-16. He worked with me in the projects involving Singlet and Triplet extension of Standard Model in the supersymmetric context.

At University of Helsinki

2011-14 I co-supervised Asli Sabanci, student of Prof. Katri Huitu, towards her Ph.D thesis during 2011-14. She worked with me in the project involving Triplet extension of Standard Model in the supersymmetric context.

Supervising Postdoc at IITH

May, 2019 I have one Postdoc from SERB Core grant. Dr. Anibarn Karan from Institute for Mathematical science has joined May, 2019 at IITH. The position is for three years. During April, 2022 he moved too IFIC, Valence, Spain as a postdoctoral fellow. Currently he is University of Calabria, Italy as a postdoc.

May 2022 Dr. Manoranjan Dutta was a postdoctoral fellow for a short time before he joined as assistant professor at North Laakhimpur college, Assam.

Supervising Master Students

So far I have supervised 9 MSc students and 1 MTech student towards their thesis.

2017-18 I supervised two Master Students: Arjun Kumar and Shilpa Jangid. The thesis works lead to a publication latter.

2018-19 I supervised Chandrima Sen towards her master thesis.

2020-21 I supervised late Pritam Roy towards his master thesis.

2021-22 I supervised Mahima Chaudhuri towards her master thesis.

2022-23 I supervised Trina Basu for one semester and Bhagya S. Babu did the full master thesis with me.

2024-25 I supervised Disha Ghosh and Gaurav Joshi towards their master thesis. Both works are expected to lead to publications.

2024-25 I supervised Jnana Sri Chand towards his MTech thesis.

Supervising BTech Students

So far I have supervised 7 EP students towards 8 projects as listed year wise.

2017-18 I supervised Mahesh Jakkapu towards his BTech Project.

2018-19 I supervised Mahesh Jakappu towards his Hons Project, which lead to a publication latter.

2018-19 I supervised Akanksha Balakrishna Dhaygude towards her BTech projects.

- 2019-20 I supervised two EP student Dhruv Gupta and Abhishek Agarwal towards their BTech projects.
- 2023 I supervised EP student Puli Reeta towards her BTech projects.
- 2025-26 I am supervising EP student Y Venkat Raghav his BTech projects.

Supervising Intern Students

So far I have supervised 6 intern students their projects as listed year wise.

- 2017 I supervised three interns Akanksha Balakrishna Dhaygude (from IITH), Shiva Vidyesh (from University of Hyderabad), Ram Purandhar S ()from IIT Madras) towards their Project.
- 2019 I supervised Ananya Janardhanan from IISER Bhopal
- 2020 I supervised Arya A from Central University Karnataka
- 2021 I supervised Aleesha KT from IISER Kolkata

Teaching at Outside IITH

- March, 2024 Lectured for Ph.D students, at Sangam School, Harish-Chandra Research Institute, Prayagraj
- August, 2022 Given lecture series for the Ph.D students at IIT Hyderabad at the Data and Machine Learning workshop by KARYashala.
- January, 2020 Given lecture series for the Ph.D students at Institute of Physics Bhubaneswar
- March, 2019 Lectured for Ph.D students, at Sangam School, Harish-Chandra Research Institute, Prayagraj
- October, 2018 Tutored for DST-SERB prep school on High Energy Physics at University of Hyderabad
- March, 2018 Lectured for Ph.D students at Sangam school, Harish-Chandra Research Institute, Prayagraj

Awards/Fellowships

- 2025 Guest Editor for The European Physical Journal Special Topics- High Energy Physics: The Path Ahead
- 2025 INR 50000 towards hosting Phoenix 2025 from The Institute of Mathematical Sciences(DAE), Chennai
- 2025 INR 3.5 Lakhs towards hosting Phoenix 2025 under ANRF's SSY grant
- 2023-24 Visiting professor at Korea Institute for Advanced Study, Seoul, South Korea
- 2023 INR 3 Lakhs towards hosting Phoenix 2023 under SERB's SSY grant
- 2023 Teachers excellence award 2023 by IIT Hyderabad
- 2023 Selected to apply for SERB's STAR proposal based on the performance of earlier projects
- 2022 SERB's Karyashla grant of 5 lakhs for "Data and Machine learning at the LHC"
- 2020 MATRICIS grant of 6.6 lakhs for three years.
- 2019 Institute of Mathematical Science, Chennai Support of INR 2 lakhs for Indo-US workshop Anomalies 2019
- 2019 Core Grant CRG/2018/004971: of INR $\sim$ 31.5 lakhs, which includes one Postdoctoral fellow and computing facilities.

- 2019 SERB Grant of INR 2 Lakhs for Anomalies 2019: SSY/2019/000196
- 2018 IUSSTF award of INR 10 lakhs for the Indo-US workshop: AUG/WS/050/2018 – Anomalies in High Energy Physics 2019
- 2016 Assistant Professor position in Indian Institute of Technology Hyderabad, India
- 2015 Young scientists award in Corfu summer school and workshop, CORFU 2015
- 2014- Offered to join as postdoctoral fellow at University of Silesia, Poland
- 2014- Offered to join in Korea University as Jeong-ho Research Fellow for 4 years
- 2014- Selected for INFN fellowship in theoretical High-energy Physics.
- 2013- Selected for Inspires faculty fellowship, a five year faculty position in India
- 2011- Adjunct Scientist in Helsinki Institute of Physics, Helsinki, Finland
- 2011- Post-doctoral researcher in University of Helsinki, Helsinki, Finland
- 2010-2011 Research Fellowship (Postdoctoral) at Korea Institute for Advanced Study, Seoul
- 2006-9 Senior Research Fellowship at Harish-Chandra Research Institute, Allahabad
- 2004-2006 Junior Research Fellowship at Harish-Chandra Research Institute, Allahabad
- 2004 NET Selected for UGC JRF in June, 2004 NET
- 2004 GATE Ranked 3rd in all India (99.89 Percentile)
- December, 2003 Selected CSIR JRF and called for SPMF examination
NET
- may-June 2003 Selected as Visiting Student Research Fellow(VSRP) student in TIFR, Mumbai
- 2002 Selected for National Scholarship for B.Sc(Physics(Hons))

Projects

- 2025 INR 50000 towards hosting Phoenix 2025 from The Institute of Mathematical Sciences(DAE), Chennai
- 2025 INR 3.5 Lakhs towards hosting Phoenix 2025 under ANRF's SSY grant
- 2023 INR 3 Lakhs towards hosting Phoenix 2023 under SERB's SSY grant
- 2022 SERB's Karyashla grant of 5 lakhs for "Data and Machine learning at the LHC"
- 2020 MATRICIS grant of 6.6 lakhs for three years.
- 2019 Institute of Mathematical Science, Chennai Support of INR 2 lakhs for Indo-US workshop Anomalies 2019
- 2019 Core Grant CRG/2018/004971: of INR $\sim$ 31.5 lakhs, which includes one Postdoctoral fellow and computing facilities.
- 2019 SERB Grant of INR 2 Lakhs for Anomalies 2019: SSY/2019/000196
- 2018 IUSSTF award of INR 10 lakhs for the Indo-US workshop: AUG/WS/050/2018 – Anomalies in High Energy Physics 2019
- 2017 Seed Grant of INR 3 Lakhs from IIT Hyderabad.

Infrastructure established at IITH

Labs, rooms, etc:

- Lab C 311 and PH320: Infrastructure of cubicles, Electrical connection, UPS, Internet connection for the Ph.D students of the Physics Department

Open area and other rooms: In the Open area and committee, conference and class rooms to enhance the discussion and create more rooms for discussions. Anti-reflecting coating to lab Ph 320, class rooms and conference room.

Computing facilities

Established a computing facility with the work stations with high committing abilities from different projects as listed below

Workstation-1	Dell Precision 5820 from HEFA
Workstation-2	Intel Core i9-9900K CPU@3.6GHz x 16, 32GB RAM, 1 TB HDD from G327 Matrics and RDF
Workstation-3	Intel Core i7-8700K CPU@3.7GHz x 12, 16 GB RAM, 1 TB HDD from RDF
Workstation-4	Intel Core i7-8700K CPU@3.7GHz x 12, 16 GB RAM, 1 TB HDD from RDF
Workstation-5	Dell, Intel Core i7-8700 CPU@3.2GHz x 8, 40 GB RAM, 1 TB HDD from G210, CRG
Workstation-6	Dell Vostro 3250 from Institute
Workstation-7	Dell Inspiron 3268 from Seed grant
Workstation-8	Apple iMac from Seed grant
APC SmartUPS	3 KVA, Online, with 6 Exide batteries from RDF
APC BackUPS x 3	1.5 KVA from Institute + RDF
DiskStation NAS	4x4TB SATA HDD from Institute
Dell Laptop	Intel Core i5, 8 GB RAM, 1 TB HDD from G327, Matrics
Dell Laptop	Intel Core 12th Gen from RDF
MacBook Pro 13	Intel Core i9, 16GB RAM, 256 GB HD from G210, CRG
MacBook Pro 14	M4, 24 GB RAM, 1 TB from Institute +RDF

Administrative Activities at IITH

DUGC

During Ph.D, 2017- I am one of the member of DUGC, physics. I was and am involved the in course management committee for EP and M.Sc. I also documented the course details for Ph.D, M.Sc and EP.

Open Day

2018-19 I was activity involved in the Physics department's open day program. Projects, design, arts, play, musics and other activity for the mental health of the students were organized. We brought eminent speakers on this occasions.

ARIIA

2019-23 I am the nodal cor-coordinator of the Atal Ranking of Institutions on Innovation and Achievement. We have been meticulously filling up the data now for a few years which is very challenging.

Faculty Shortlisting

2016-24 I was involved in calculating the API score for the short listing of the applications for the faculty position at IIT Hyderabad.

Ph.D Thesis Committee

2016- I am in the thesis committee of several students of other faculties. I have been involved in their annual evaluation, RPS and comprehensive exams.

Ph.D Selection Committee

2016- I am in the selection committee which sets the written question papers and conduct the interviews. I have been doing this duty since I joined.

Faculty advisor

2019-22 I have been the faculty advisor of EP2019 batch and guide and help them whenever they needed.

Ph.D Student Lab

2019- With the help of the HPD and the department I was actively involved in building the C311 Lab for the theory student. In the new physics building I have been involved with the infrastructure development of PH320.

Departmental Infrastructure

2020- I have been supervising the enhancement of the academic facilities of the lecture rooms, commitment rooms, open area and other Labs of our department.

Coordinating the seminar speakers/visitors

2017- I have been involved as one of the active volunteers in coordinating departmental seminar speakers, adjunct faculties and other visitors.

Organising Workshops/Conferences

2018- I have been involved in organising various national and international level workshops/conferences at IITH.

GATE and JEE duties

2017- I have done my bit organizing these national level exam whenever asked, being in the city or out stationed.

Science Outreach

Seminars	Given a seminar promoting science at Sangareddy Government College
Guiding interns	Taught and worked with various in interns from various institutes. Some of them has done undergrad or post graduate thesis with me.
Open days	I am actively involved in organising open days promoting science activities.
Pi day	I am actively involved in organising Pi day promoting science activities.
Science quizzes	Organised science quizzes from local local and college students
Popular article writing	:I was also involved in IITH Magazine Lexicon writing scientific articles.

Current International collaborations

- 1 Prof. Eung Jin Chun, Korea Institute for Advanced Study, Seoul, South Korea
- 2 Prof. Mariana Frank Concordia University, Canada
- 3 Prof. Luigi Della Rosa University of Calabria, INFN Calabria, Italy
- 4 Prof. Yuji Omura, Kindai University, Japan
- 5 Prof. Jeonghyeon Song, Konkuk University, South Korea

Current National collaborations

- 1 Prof. Kirtiman Gosh, Institute of Physics, Bhubaneswar
- 2 Prof. Manimal Mitra, Institute of Physics, Bhubaneswar
- 3 Dr. Rusa Mandal, IIT Gandhinagar
- 4 Students, Postdoc, faculties at IITH

Computer Skills

Operating Systems	Unix, Linux, Windows
Platforms	Solaris, SuSe, Redhat, Fedora, Ubuntu, Windows 2000/XP
Languages	Fortran, C++, C, Mathematica, LaTeX, HTML, Unix & Linux Shell scripting
Physics Soft-wares	PYTHIA, Prospino, ISAJET, CalcHEP, CompHEP, micrOMEGAs CPsuperH, Alpgen, PYTHIA-Calcchep interface and PYTHIA-TAUOLA interface
Calcchep-PYTHIA	Interface has been used in my study of CP-violating Supersymmetry. Some UED couplings are also introduced in Calcchep-model file and extensive simulation has been done for UED via Calcchep-PYTHIA interfacing and mixing of generated events. Type III and Type I seesaw models are also implemented in Calcchep in current projects.

Personal Data

Full Name	Priyotosh Bandyopadhyay
Date of birth	29.12.1981
Gender	Male
Marital Status	Married
Blood Type	B^+
Nationality	Indian
Home Address	Barabagan, 1-Palli, Suri, Dist-Birbhum, West Bengal, PIN-731103, Phone: +91-3462-259-177

Languages

Bengali	1st language (Mother tongue); can read, speak, and write
English	2nd language; can read, speak, and write
Hindi	3rd language; can read, speak, and write
Korean	Read Only

Contact Address

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Extra Curricular Activities

Recently I engage in writing and started writing in our institute magazine Lexicon. I love running. I participated in few marathons and completed successfully. I am involved in physical activity. I also love reading, watching sports and movies. Photography is my other hobby. I was involved in child education for those who can not afford to go to school during my stay in India. I would love to be involved in such activity in more meaningful way in future.

List of Referees

Prof.Claudio Coriano

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List of Publications in refereed journals

- 1 Dark clouds to silver linings over the hyperchargeless scalar triplets, Priyotosh Bandyopadhyay, Snehashis Parashar, [PRD 112 \(2025\) 5, 055032](#), IF=4.6
- 2 Fate of the scalar quartic couplings in the inert models, Priyotosh Bandyopadhyay, Pram Milan Robin, [PRD \(2025\)](#) , IF=4.6
- 3 Probing a scalar singlet-triplet extension of the standard model via vector boson fusion at a muon collider, Priyotosh Bandyopadhyay, Snehashis Parashar, [PRD 110 \(2024\) 11, 115032](#), IF=4.6
- 4 Probing Inert Triplet Model at a multi-TeV muon collider via vector boson fusion with forward muon tagging, Priyotosh Bandyopadhyay, Snehashis Parashar, Chandrima Sen, Jeonghyeon Song, [JHEP 07 \(2024\) 253](#), IF=5.0
- 5 Interplay of inert doublet and vector-like lepton triplet with displaced vertices at the LHC/FCC and MATHUSLA, Priyotosh Bandyopadhyay, Snehashis Parashar, Chandrima Sen, Marriaa Frank, [JHEP 03 \(2024\) 109](#) , IF=5.0
- 6 Discerning Singlet and Triplet scalars at the electroweak phase transition and Gravitational Wave, Priyotosh Bandyopadhyay, Shilpa Jangid, [PRD 107 \(2023\) 5, 055032](#), IF=4.6
- 7 Boosted displaced decay of right-handed neutrinos at CMS, ATLAS and MATHUSLA, Priyotosh Bandyopadhyay, Eung Jin Chun, Chandrima Sen, [JHEP 02 \(2023\) 103](#)
- 7 Semi-Annihilation of Fermionic Dark Matter, Priyotosh Bandyopadhyay, Debajyoti Choudhury, Divya Sachdeva, [Phys.Rev.D 107 \(2023\) 1, 015020](#), IF=4.6

- 9 Phenomenology of Scalar Leptoquarks at the LHC in Explaining the Radiative Neutrino Mass, Muon $g-2$ and Lepton Flavour Violating Observables, Snehashis Parashar, Anirban Karan, Avnish, Priyotosh Bandyopadhyay, Kirtiman Ghosh, [Phys.Rev.D 106 \(2022\)9, 095040](#) , IF=4.6
- 10 Distinguishing signatures of scalar leptoquarks at hadron and muon colliders, Priyotosh Bandyopadhyay, Anirban Karan, Rusa Mandal, Snehashis Parashar, [Eur.Phys.J.C 82 \(2022\) 10, 916](#), 41 pages, IF=4.8
- 11 Constraining scalar doublet and triplet leptoquarks with vacuum stability and perturbativity, Priyotosh Bandyopadhyay, Shilpa Jangid, Anirban Karan, [Eur. Phys.J.C 82 \(2022\)516](#), 44 pages, IF=4.8
- 12 Secluded Dark Matter in Gauged $B - L$ model, Priyotosh Bandyopadhyay, Manimala Mitra, Rojalin Padhan, Abhishek Roy, Michael Spannowsky, [JHEP 05\(2022\)182](#), 40 pages, IF=5.0
- 13 Displaced Higgs production in Type-III Seesaw at the LHC/FCC, MATHUSLA and Muon collider, Chandrima Sen, Priyotosh Bandyopadhyay, Saunak Dutta, Aleesha KT, [Eur.Phys.J.C 82 \(2022\) 3, 230](#), IF=4.8
- 14 Distinguishing Leptoquarks at the LHC/FCC, Priyotosh Bandyopadhyay, Saunak Dutta, Anirban Karan, Mahesh Jakkapu, [Nucl.Phys.B 971 \(2021\) 115524](#), IF=3.16
- 15 Relativistic Freeze-in with Scalar Dark Matter in a Gauged B-L Model and Electroweak Symmetry Breaking, Priyotosh Bandyopadhyay, Manimala Mitra, Abhishek Roy, [JHEP05\(2021\)150](#), IF=5.0
- 16 Zeros of Amplitude in the Associated Production of Photon and Leptoquark at $e-p$ Collider, Priyotosh Bandyopadhyay, Saunak Dutta, Anirban Karan, [Eur.Phys.J.C 81 \(2021\) 4, 315](#), 26 pages, IF=4.8
- 17 Obscure Higgs boson at Colliders, Priyotosh Bandyopadhyay, Antonio Costantini, 16 pages, [Phys.Rev.D 103 \(2021\) 015025](#), IF=4.6
- 18 Scrutinizing Vacuum Stability in IDM with Type-III Inverse seesaw, Priyotosh Bandyopadhyay, Shilpa Jangid, Manimala Mitra, [JHEP02\(2021\)075](#), IF=5.0
- 19 Vacuum Stability in Inert Higgs Doublet Model with Right-handed Neutrinos, Priyotosh Bandyopadhyay, Bhupal Dev, Shilpa Jangid, Arjun Kumar, [JHEP 08 \(2020\) 154](#), 36 pages, JHEP, IF=5.0
- 20 Distinguishing Inert Higgs Doublet and Inert Triplet Scenarios Shilpa Jangid, Priyotosh Bandyopadhyay, [Eur.Phys.J.C 80 \(2020\) 8, 715](#), 34 pages, IF=4.8
- 21 Feeble neutrino portal dark matter at neutrino detectors, Priyotosh Bandyopadhyay, Eung Jin Chun, Rusa Mandal, [JCAP08\(2020\)019](#), IF=5.5
- 22 Investigating the Production of Leptoquarks by Means of Zeros of Amplitude at Photon Electron Collider, Priyotosh Bandyopadhyay, Saunak Dutta, Anirban Karan, [Eur.Phys.J.C 80 \(2020\) 6, 573](#), 43 pages, IF=4.8
- 23 Phenomenology of Higgs bosons in inverse seesaw model with Type-X two Higgs doublet at the LHC, Priyotosh Bandyopadhyay, Eung Jin Chun, Rusa Mandal, 22 pages, [JHEP08\(2019\)169](#)., IF=5.0
- 24 Long-lived triplinos and displaced lepton signals at the LHC, Asli Sabanci Keceli, Priyotosh Bandyopadhyay, Katri Huitu, [Eur.Phys.J.C 79 \(2019\) 4, 345](#), IF=4.3

- 25 Scrutinizing Right-Handed Neutrino Portal Dark Matter With Yukawa Effect, Priyotosh Bandyopadhyay, Eung Jin Chun, Rusa Mandal, [Phys.Lett.B 788 \(2019\) 530-534](#), 5 pages, IF=5.02
- 26 Revisiting scalar leptoquark at the LHC , Priyotosh Bandyopadhyay, Rusa Mandal, [Eur.Phys.J.C 78 \(2018\) 491](#) 14 pages, IF=4.8
- 27 Scalar Dark Matter in Leptophilic Two-Higgs-Doublet Model, Priyotosh Bandyopadhyay, Eung Jin Chun, Rusa Mandal, [Phys.Lett.B 779 \(2018\) 201-205](#) , 5 pages, IF=5.02
- 28 Distinguishing charged Higgs bosons from different representations at the LHC Priyotosh Bandyopadhyay, Antonio Costantini, [JHEP 1801 \(2018\) 067](#), 29 pages, IF=5.0
- 29 Implications of right-handed neutrinos in B-L extended standard model with scalar dark matter , Priyotosh Bandyopadhyay, Eung Jin Chun, Rusa Mandal, [Phys.Rev.D 97 \(2018\) 1, 015001](#), 12 pages, IF=4.6
- 30 Displaced lepton flavour violating signatures of right-handed sneutrinos in $U(1)'$ supersymmetric models, Priyotosh Bandyopadhyay, [JHEP 1709 \(2017\) 052](#) , 17 pages, IF=5.0
- 31 Vacuum stability in an extended standard model with a leptoquark, Priyotosh Bandyopadhyay, Rusa Mandal, arXiv:1609.03561[hep-ph], [Phys.Rev. D95 \(2017\) no.3, 035007](#), 14 pages (NUMBER OF CO-AUTHORS-1, 14 pages, IF=4.6)
- 32 General analysis of the charged Higgs sector of the $Y=0$ triplet-singlet extension of the MSSM at the LHC, Priyotosh Bandyopadhyay, Claudio Coriano, Antonio Costantini, [Phys. Rev. D 94\(2016\), 055030](#), 26 pages (NUMBER OF CO-AUTHORS-2, IF=4.6)
- 33 Bounds on the Conformal Scale of a Minimally Coupled Dilaton and Multi-Leptonic Signatures at the LHC, Priyotosh Bandyopadhyay, Antonio Constantini, Claudio Coriano and Luigi Delle Rose, [JHEP 1609 \(2016\) 084](#), 23 pages (NUMBER OF CO-AUTHORS-3, IF=5.0)
- 34 Non-standard charged Higgs decay at the LHC in Next-to-Minimal Supersymmetric Standard Model, Priyotosh Bandyopadhyay, Katri Huitu and Saurabh Niyogi, [JHEP 1607 \(2016\) 015](#), 24 pages (NUMBER OF CO-AUTHORS-2, IF=5.0)
- 35 Probing the Hidden Higgs bosons of the $Y = 0$ Triplet- and singlet -Extended Supersymmetric Standard Model at the LHC, Priyotosh Bandyopadhyay, Antonio Constantini, Claudio Coriano, accepted in [JHEP12\(2015\)127](#), 29 pages (NUMBER OF CO-AUTHORS-2, IF=5.0)
- 36 Perspectives on a Supersymmetric Extension of the Standard Model with a $Y=0$ Higgs Triplet and a Singlet at the LHC, Priyotosh Bandyopadhyay, Claudio Coriano, Antonio Costantini, [JHEP09\(2015\)045](#) (publisher SpringerLink, ISSN:1029-8479), 40 pages (NUMBER OF CO-AUTHORS-2 IF=5.0)
- 37 Lepton flavour violating signature in supersymmetric $U(1)'$ seesaw models at the LHC, Priyotosh Bandyopadhyay, Eung Jin Chun. [JHEP05\(2015\)045](#) (publisher SpringerLink, ISSN:1029-8479), 16 pages (NUMBER OF CO-AUTHORS-1, IF=5.0)
- 38 Multi-Lepton Signatures of the Triplet Like Charged Higgs at the LHC, Priyotosh Bandyopadhyay, Katri Huitu, Asli Sabanci Keceli, [JHEP05\(2015\)026](#) (publisher SpringerLink, ISSN:1029-8479), 25 pages (NUMBER OF CO-AUTHORS-2, IF=5.0)

- 39 Naturality vs perturbativity, $B_s \rightarrow X_s \gamma$, and LHC data in triplet extension of MSSM, Priyotosh Bandyopadhyay, Stefano Di Chiara, Katri Huitu, Asli Sabanci Keçeli, [JHEP 1411 \(2014\) 062](#) (publisher SpringerLink, ISSN:1029-8479), 28 pages (NUMBER OF CO-AUTHORS-3, IF=5.0)
- 40 Production of two Higgses at the Large Hadron Collider in CP-violating supersymmetry:, Priyotosh Bandyopadhyay, Katri Huitu, [JHEP 1311 \(2013\) 058](#) (publisher SpringerLink, ISSN:1029-8479), 22 pages (NUMBER OF CO-AUTHORS-1, IF=5.0)
- 41 Status of $Y=0$ Triplet Higgs with supersymmetry in the light of ~ 125 GeV Higgs discovery, Priyotosh Bandyopadhyay, Katri Huitu, Asli Sabanci, [JHEP 1310 \(2013\) 091](#) (publisher SpringerLink, ISSN:1029-8479), 24 pages, (NUMBER OF CO-AUTHORS-2, IF=5.05)
- 42 Higgs Signatures in Inverse Seesaw Model at the LHC, Priyotosh Bandyopadhyay, Eung Jin Chun, Hiroshi Okada, Jong-Chul Park, [JHEP 1301 \(2013\) 079](#) (publisher SpringerLink, ISSN:1029-8479), 25 pages (NUMBER OF CO-AUTHORS-3, IF=5.05)
- 43 Probing Higgs in Type III Seesaw at the LHC, Priyotosh Bandyopadhyay, Eung Jin Chun and Suyong Choi and Kyungnam Min, [PhysRevD.85.073013](#) (publisher APS, ISSN 1550-2368 (online), 1550-7998 (print)), 5 pages, (NUMBER OF CO-AUTHORS-3, IF=4.6)
- 44 Right-handed sneutrino dark matter in $U(1)'$ seesaw models and its signatures at the LHC, Priyotosh Bandyopadhyay, Eung Jin Chun, Jong-Chul Park, [JHEP 06 \(2011\) 129](#) (publisher SpringerLink, ISSN:1029-8479), 26 pages (NUMBER OF CO-AUTHORS-2, IF=5.0)
- 45 An unusual signal of Higgs boson in supersymmetry at the LHC, Priyotosh Bandyopadhyay, Pradipta Ghosh, Sourov Roy, [Phys.Rev. D84 \(2011\) 115022](#) (publisher APS, ISSN 1550-2368 (online), 1550-7998 (print)), 7 pages, (NUMBER OF CO-AUTHORS-2, IF=5.51), IF=4.6
- 46 Boosted top quarks in supersymmetric cascade decays at the LHC, Priyotosh Bandyopadhyay, Biplob Bhattacharjee, [Phys. Rev. D 84, 035020, 2011](#) (publisher APS,ISSN 1550-2368 (online), 1550-7998 (print)), 9 pages, (NUMBER OF CO-AUTHORS-1, IF=4.6)
- 47 Higgs production in CP-violating supersymmetric cascade decays: probing the 'open hole' at the Large Hadron Collider, Priyotosh Bandyopadhyay, [JHEP 08 \(2011\) 016](#) (publisher SpringerLink, ISSN:1029-8479), 22 pages (NUMBER OF CO-AUTHORS-0, IF=5.0)
- 48 Displaced Higgs production in type III seesaw, Priyotosh Bandyopadhyay, Eung Jin Chun, [JHEP 1011 \(2010\) 006](#), (publisher SpringerLink, ISSN:1029-8479,), 12 pages (NUMBER OF CO-AUTHORS-1, IF=5.0)
- 49 Search for Higgs bosons of the Universal Extra Dimensions at the Large Hadron Collider, Priyotosh Bandyopadhyay, Biplob Bhtayacharya, Asesh Krishna Datta, [JHEP 03 \(2010\) 048](#), (publisher SpringerLink, ISSN:1029-8479), 24 pages (NUMBER OF CO-AUTHORS-2, IF=5.0)

- 50 Two Higgs Doublet Type III Seesaw, Priyotosh Bandyopadhyay, mu-tau symmetry at LHC, Priyotosh Bandyopadhyay, Sandhya Chaubey, Manimala Mitra, [JHEP 10 \(2009\) 012](#) (publisher IOPscience, ISSN: 1029-8479), 51 pages (NUMBER OF CO-AUTHORS-2, IF=5.0)
- 51 Probing non-universal gaugino masses via Higgs boson production under SUSY cascades at the LHC: A Detailed study, Priyotosh Bandyopadhyay, [JHEP 07 \(2009\) 102](#) (publisher IOPscience, ISSN: 1029-8479), 44 pages, IF=5.0
- 52 Signatures of gaugino mass non-universality in cascade Higgs production at the LHC, Priyotosh Bandyopadhyay, Aresh Krishna Datta, Biswarup Mukhopadhyaya, [Physics Letters B 670 \(2008\) 5–11](#) (publisher ScienceDirect, ISSN: 0370-2693), arXiv:0806.2367 [hep-ph], 7 pages (NUMBER OF CO-AUTHORS-2, IF=5.02)
- 53 Associated Higgs production in CP-violating supersymmetry: Probing the 'open hole' at the large hadron collider, Priyotosh Bandyopadhyay, Amitava Datta, Aresh Krishna Datta, Biswarup Mukhopadhyaya, [Phys.Rev.D 78 \(2008\) 015017](#) (publisher APS, ISSN 1550-2368 (online), 1550-7998 (print)), 11 pages (NUMBER OF CO-AUTHORS-3, IF=4.6)

List of articles in arXiv/Journals

- 1 Probing Discerning Signatures of SM and Various Seesaw Models at e^-e^- Collider, Priyotosh Bandyopadhyay, Anirban Karan, Chandrima Sen, 2011.04191 [hep-ph]

List of Publications in conference proceedings

- 1 Deciphering the Secrets of the Long-Lived Particles at the Colliders, Chandrima Sen, Priyotosh Bandyopadhyay, Mariana Frank, Snehashis Parashar., [PoS ICHEP2024 \(2025\) 280](#)
- 2 Signatures of the Inert Triplet Model from Vector Boson Fusion at a Muon Collider, Snehashis Parashar, Priyotosh Bandyopadhyay, Chandrima Sen, Jeonghyeon Song, [PoS ICHEP2024 \(2025\) 305](#)
- 3 Phenomenology of Scalar Leptoquarks: Neutrino Mass, $g - 2$, and B-Anomalies, Snehashis Parashar, Priyotosh Bandyopadhyay, Anirban Karan, Rusa Mandal, Avnish et al., [Proc.Phys. 304 \(2024\) 919-921](#)
- 4 On the Fate of Electroweak Vacuum and Order of Phase Transition in Beyond Standard Model Scenarios, Priyotosh Bandyopadhyay, Shilpa Jangid, [Proc.Phys. 304 \(2024\) 263-268](#)
- 5 Searching for Scalar Leptoquarks at the LHC/FCC and a Muon Collider, Snehashis Parashar, Priyotosh Bandyopadhyay, Anirban Karan, Rusa Mandal, Avnish Avnish et al., [PoS LHCP2023 \(2024\) 240](#)
- 6 Distinguishing Different BSM Signatures at Present and Future Colliders, Saunak Dutta, 2)Priyotosh Bandyopadhyay, and Anirban Karan, [Andromeda Conference ProceedingBSM-2021.11, 9](#)

- 7 Charged Higgs in Beyond the Minimal Supersymmetric Standard Model at the LHC, Katri Huitu, P. Bandyopadhyay, S. Dichiara, M. Frank, D.K. Ghosh, A.S. Keceli, S. Niyogi, S.K. Rai, I. Saha, H. Waltari, [PoS CHARGED2016 \(2017\) 023](#), 5 pages, IF=0.63
- 8 Charged Higgs bosons in the extended supersymmetric scenario at the LHC, Priyotosh Bandyopadhyay, [PoS LHCP2016 \(2016\) 080](#), 9 pages, IF=0.63
- 9 Extended Higgs sectors in the context of supersymmetry at the LHC, Priyotosh Bandyopadhyay, [EPJ Web Con. 129 \(2016\) 00009](#)
- 10 Higgs bosons: discovered and hidden, in extended Supersymmetric Standard Models at the LHC, Priyotosh Bandyopadhyay, Antonio Constantini, Claudio Coriano, [PoS CORFU2015 \(2016\) 069](#), 12 pages, IF=0.63
- 11 Triplet Extended MSSM: Fine Tuning vs Perturbativity and Experiment Priyotosh Bandyopadhyay, Stefano Di Chiara, Katri Huitu, Aslı Sabancı Keçeli, C14-07-02, [Nucl.Part.Phys.Proc. 273-275 \(2016\) 595-601](#), contribution to ICHEP 2014 conference proceeding, 7 pages, IF=0.134
- 12 Higgs(es) in triplet extended supersymmetric standard model at the LHC Priyotosh Bandyopadhyay, Katri Huitu, Aslı Sabancı Keçeli, C14-07-02, [Nucl.Part.Phys.Proc. 273-275 \(2016\) 2473-2475](#), contribution to ICHEP 2014 conference proceeding, 3 pages, IF=0.134
- 13 Working Group Report: Physics at the Large Hadron Collider at WHEPP 2010, [Pramana 76 \(2011\) 707-723](#) (publisher IAS, ISSN:0304-4289, 17 pages, IF=0.562)

Organizing Workshops/Conference at IIT Hyderabad

- | | |
|-----------------|--|
| 9-11/07/2025 | An international conference, Phoenix 2025 at IIT Hyderabad |
| 18-20/12/2023 | An international conference, Phoenix 2023 at IIT Hyderabad |
| 22-28/08/2022 | A DST Karyashala workshop: Data and Machine Learning at the LHC, IIT Hyderabad |
| | |
| 14/03/2022 | Pi-Day, IIT Hyderabad |
| 10-12/11/2021 | Anomalies 2021 at IIT Hyderabad |
| 05/2020-03/2021 | HEP seminar series (12 lectures) throughout the year at IIT Hyderabad |
| 11-13/09/2020 | 'Anomalies 2020' at IIT Hyderabad |
| 21-24/07/2019 | Activity week post 'Anomalies 2019' at IIT Hyderabad |
| 13-19/10/2019 | TEQIP workshop 'QuantOEM' at IIT Hyderabad |

National and International visitors Invited at IIT Hyderabad

- | | |
|---------------|---|
| 2017-December | Prof. Eung Jin Chun from KIAS, South Korea and Dr. Kenji Nishiwaki from KIAS, South Korea visited for seminars at the department of Physics, IITH |
|---------------|---|

2018-July	Prof. Mustansir Barma from TIFR Hyderabad visited us for a seminar in Open day, 2018.
2018-July	Prof. Pravabati Chingangbam from IIA Bangalore visited us for a seminar
2019-July	Prof. Eung Jin Chun visited prior to Anomalies , 2019 for research collaboration and seminar
2023,Dec-24,Jan	Prof. Eung Jin Chun visited as Adjunct faculty of IITH for lecture series
July, 2024	Dr. Joyita Lahiri. from University of Hamburg for a seminar.
2024-October	Prof. Eunji Chun visited as Adjunct faculty of IITH for lecture series
September, 2025	Prof. Prasenjit Sen from Harish-Chandra Research Institute for a seminar.

Conferences/Schools/Workshops Attended

9-11/07/25	Phoenix 2025 at IIT Hyderabad
6-9/03/25	Flavor Physics Week at IIT Gandhinagar
30/12/24- 6/01/25	Hearing beyond the standard model with cosmic sources of Gravitational Waves at ICTS, Bangalore
14-18/10/2024	Belle analysisy at IIT Hyderabad.
14-18/10/2024	Particle Physics and Cosmology at IIT Hyderabad.
19-14/06/2024	SUSY2024, IFT, Madrid, Spain
March,2024	Sangam School at Harish-Chandra Research Institute, Prayagraj
18-20/12/2023	Phoenix 2023 at IIT Hyderabad.
17-21/07/2023	The 2nd workshop on Symmetry and Structure formation of the universe, at Jeonbuk National University, Jeonju. South Korea
03-14/07/2023	QUC summer school on A.I in High Energy Physics, KIAS, Seoul
24-25/03/2023	HHPS 2023 at IIT Gandhinagar
21-22/03/2023	Workshop on Particle Physics, Astrophysics and Cosmology (WPAC 2023) 2023 at Shiv Nadar University, Noida
15-22/02/2023	IMHEP, Institute of Physics, Bhubaneswar
13-18/11/2022	The 2nd AEI workshop and 10th KIAS workshop on particle physics and cosmology
30/09- 04/10/2022	Beyond Standard Model Scenarios, IOP Bhubaneswar, India
22-28/08/2022	Data and Machine learning at the LHC, IIT Hyderabad, India
31/08- 12/09/2021, Online	VIRTUAL SCHOOL ON FLAVOR STRUCTURE OF THE STANDARD MODEL
23-28/08/2021, Online	The XXVIII International Conference on Supersymmetry and Unification of Fundamental Interactions (SUSY 2021)
24-26/05/2021, Online	Phenomenology 2021 Symposium, University of Pittsburgh
01-03/03/2021, Online	Unveiling Hidden Physics Beyond the Standard Model at the LHC
09-13/11/2020, Online	Less Travelled Path of Dark Matter: Axions and Primordial Black Holes (ONLINE)

11-13/09/2020, Online	Anomalies 2020 at IITH
02-10/03/2019	Activity Week at HRI, Prayagraj INDIA: Attended and given a seminar.
12-17/01/2019	IMHEP 2019, IOP, Bhubaneswar, INDIA: Attended and given a seminar at the parallel session.
20/10- 01/11/2018	8th KIAS workshop, Seoul: Attended and given a seminar at the parallel session.
7-9/05/2018	Pheno 2018, Pittsburgh, USA: Attended Pheno 2018 at Pittsburgh and given a seminar at the parallel session.
05-09/03/2018	Sangam@HRI: Attended and gave a seminar at Sangam@HRI, Allahabad, India
27-30/01/2018	Top@IISERK: Attended and given a seminar at Top@IISERK, India
06-09/11, 2017	7 th KIAS phenomenological workshop, KIAS, Seoul, South Korea
16-18/11, 2016	Understanding the origin of the invisible sector, at University of Hyderabad, India
27-30/06, 2016	QCD@Work: International Workshop on QCD Theory and Experiment 2016, Martina Franka, Italy
13-18/06, 2016	Fourth Annual Large Hadron Collider Physics Conference (LHCP2016), Lund, Sweden
15-17/02, 2016	KIAS/KNRC School on Neutrino Physics, KIAS, South Korea
30/01-04/02, 2016	Workshop on Particle physics, String theory and Cosmology, High1 Resort, South Korea
Spetember, 01-11, 2015	Summer School and Workshop on the Standard Model and Beyond, Corfu, Greece
28th November, 2014	One day workshop at INFN Bari, Bari, Italy
26th October, 2014	Particle Physics Day, Jyväskylä, Finland
July, 02-09, 2014	ICHEP 2014, Valencia, Spain.
March, 11-13, 2014	Physics Days 2014, Tampere, Finland.
January, 2-7, 2014	Spaatind 2014, Skeikampen, Norway.
October, 24, 2013	Physics Day, Helsinki Institute of Physics, Helsinki, Finland.
August, 25-31, 2013	SUSY 2013, ICTP, Italy.
June, 4-7, 2013	Cosmological perturbations post-Planck, University of Helsinki, Finland
March, 14-16, 2013	Physics Day Conference, Espoo, Finland
February, 25-27, 2013	HiggStop-2013, BITs, Goa, India
September, 10-14, 2012	2nd KIAS Phenomenological Workshop, KIAS, Seoul, South Korea
August, 12-18, 2012	SUSY 2012, Peking University, Beijing, China
November, 23-25, 2011	GUT and Leptogenesis, University of Calcutta, Kolkata, India

November, 17-19, 2011	KIAS phenomenology workshop, KIAS, Seoul, Korea
November, 13-16, 2011	Theoretical Issues in Neutrino Physics, KIAS, Seoul, Korea
November, 09-12, 2011	6th International Workshop on Low Energy Neutrino Physics, SNU, Seoul, Korea
October, 24-29, 2011	KIAS School on MadGraph for LHC Physics, KIAS, Seoul, Korea
September, 05-10, 2011	IPMU-YITP School and Workshop on Monte Carlo Tools for LHC, YITP, Kyoto University, Japan.
August, 9 - 11, 2011	APCTP 2011 LHC Physics Workshop, Konkuk University, Seoul, Korea.
January, 04-08, 2011	Dark Matter in the LHC Era: Direct and Indirect Searches, Kolkata, India
December, 2010	21st Busan Particle Physics Workshop, Busan, Korea
September-October, 27-01, 2010	COSMO/CosPA 2010, University of Tokyo, Tokyo, Japan
August, 23-28, 2010	SUSY 10, University of Bonn, Bonn, Germany
August, 10 - 12, 2010	APCTP 2011 LHC Physics Workshop, Konkuk University, Seoul, Korea.
March, 26-30, 2010	International Linear Collider Workshop 2010, Beijing, China
Jan, 2-12, 2010	WORKSHOP ON HIGH ENERGY PHYSICS PHENOMENOLOGY (WHEPP XI), PRL, Ahmedabad, India
June-July, 29/06/09-02/07/09	Conference on Topical Issues in LHC Physics, Trieste, Italy
June, 15/06/09-26/06/09	Summer School on Particle Physics in the LHC Era, Trieste, Italy
February, 16-20, 2009	Getting Ready for Physics at the LHC, HRI, Allahabad, India
January, 15-17, 2009	BSMLHC-2009, Topical Meeting on "Beyond the Standard Model Physics at the LHC", Indian Association for the Cultivation of Science, Kolkata, India
January, 07/01-09/01, 2009	Nu Horizons '09, HRI, Allahabad, India
December, 19-24, 2008	International Workshop on LHC, IISER, Kolkata, India
Dec, 08 -Jan, 09	Particle Physics at the age of LHC, Israel, selected
September, 21/09-04/10, 2008	LHC Workshop at HRI, Allahabd, India
August, 2008	3rd CERN-Fermilab Hadron Collider School at Fermilab, Batavia, USA
February, 2008	Nuhorizone, HRI, Allahabad, India
December, 2007	Strings To LHC, Fireflies, Bangalore, India

November, 2007	QCD school at the LHC, Harish-Chandra Research Institute, Allahabad, India
January, 2007	WIN07, SINP, Kolkata, India
December, 2006	Topical Meeting on physics at the LHC, HRI, Allahabad, India
September, 2006	LHC workshop, TIFR, Mumbai, India
February 2006	XIX SERC Main School in Theoretical High Energy Physics, PRL, Ahmedabad, India

Collaborative Visits

20-27/07/2022	RKMVU, Belur India
17-26/06/2025	The Institute for Mathematical Sciences, Chennai India
8-14/12/2024	Institute of Physics, Bhubaneswar India
17-30/05/2024	University of Helsinki and Helsinki Institute of Physics. Finland
31/05-7/06/2024	University of Valencia and IFIC Valencia, Spain
7-13/02/2024	IACS, Kolkata India
10/05-15/08/2023	Korea Institue for Advanced Study, Seoul, South Korea as visiting professor
30/09-06/10/2022	Institute of Physics Bhubaneswar, India
04-21/07/2022	University of Montreal, Canada
22-27/07/2022	University of Pittsburg, USA
12-18/03/2022	Saha Institute of Nuclear Physics, India
13-18/02/2020	Institute of Physics Bhubaneswar, India
23/12/19-04/01/2020	Kindai University, Osaka, Japan
14-23/12/2019	KIAS, Seoul, South Korea
02-10/03/2019	HRI, Prayagraj, India
12-18/01/2019	Institute of Physics Bhubaneswar, India
10-29/12/2018	INFN Lecce, Italy
20/10-01/11/2018	KIAS, Seoul, South Korea
7-9/05/2018	University of Pittsburgh and given a seminar
01-2/05/2018	University of Washington at St. Louise and given a seminar
22/06-10/07/2017	Institute of Mathematical Sciences, Chennai, India
04/11-03/12/2017	Korea Institute of Advanced Study, Seoul, South Korea
24/06-05/07/2017	Institute of Mathematical Sciences, Chennai, India
18/06-23/06/2016	University of Helsinki, Helsinki, Finland
22/02-29/03/2016	Institute of Mathematical Sciences, Chennai, India

26/01-20/02/2016	Korea Institute of Advanced Study, Seoul, South Korea
13/06-23/06/2015	University of Helsinki, Helsinki, Finland
07/09-20/09/2014	Korea Institute of Advanced Study, Seoul, South Korea
10/11-13/11/2013	DESY, Hamburg, Germany
13/11-16/11/2013	KITP, Kralruhe, Germany
28/11-03/12/2012	INFN Padova, Padova, Italy
07/07-20/07/2012	Harish-Chandra Research Institute, Allahabad
05/06-18/06/2011	University of Helsinki, Helsinki, Finland
29/05-25/06/2010	University of Helsinki, Helsinki, Finland
25/06-27/06/2010	Harish-Chandra Research Institute, Allahabad

Talks Given in Conferences and Seminars

21/07/2025	Our existence and Higgs boson(s)?, RKMVU, Belur, Howrah
19/06/2025	Fate of scalar quartic couplings in the inert models , IMSc, Chennai
6-9/03/2025	Fixed points in the inert models and enhancing vacuum stability for heavy fermionic extension, Flavour Physics week II, IIT Gandhinagar
27/02/2025	Fate of the fixed points of the quartic couplings in the Inert models, Harish-Chandra Research Institute , Prayagraj
12/12/2024	Perspective of Leptoquarks, Institute of Physics Bhubaneswar
13-19/10/2024	Some about Leptoquarks, Belle Analysis workshop at IIT Hyderabad
10-14/06/2024	Interplay of inert doublet and vector-like lepton triplet with displaced vertices at the LHC/FCC and MATHUSLA, SUSY2024, IFT, Madrid, Spain
03/06/2023	Displaced Heavy leptons at the colliders,IFIC, Valencia, Spain
28/05/2024	Displaced Heavy Leptos at Collider,University of Helsinki, Helsinki, Finland
07/08/2023	Stability, Unitarity and FOPT in the presence of neutrino, Higgs and dark matter, Yonsei university, Seoul, South Korea
28/07/2023	Interplay of Higgs boson, Dark matter and Neutrino, Asia Pacific Centre for Theoretical Physics, Pohang, South Korea
17/07/2023	Perspective of Leptoquark at different colliders, The 2nd workshop on Symmetry and Structure formation of the universe, at Jeonbuk National University, Jeonju. South Korea
02/06/2023	Perspective of Leptoquark at different colliders, Korea Institute for Advanced Study, Seoul
24/03/2023	Perspective of Leptoquark at different colliders, HHHPS 2023, IIT Gandhinagar, India, plenary talk

- 22/03/2023 Perspective of Leptoquark at different colliders, WPAC 2023, Shiv Nadar University, Noida, plenary talk
- 15-22/02/2023 Interplay of Higgs boson, Dark matter and Neutrino, IMHEP 2023, Bhubaneswar, India, plenary talk
- 13-18/11/2022 Interplay of Higgs boson, Dark matter and Neutrino, The 2nd AEI workshop and 10th KIAS workshop on particle physics and cosmology, Jeju, South Korea plenary talk
- 26/07/2022 Radiation amplitude zero, University of Pittsburgh, USA
- 14/03/2022 Public Lecture on Pi-day at IIT Hyderabad
- 30/12/2021 From the cosmos to the atoms and more , Invited colloquium at North Lakhimpur College, Assam, India
- 26/08/2021 Charged Higgs from different representations at the LHC, SUSY 2021
- 24/05/2021 Distinguishing signatures of scalar leptoquarks at the LHC and Muon colliders, SUSY 2021
- 25/05/2021 Triplet Charged Higgs bosons at the LHC at Pheno 2021, University of Pittsburgh
- 06/07/2020 Vacuum stability of Extended Higgs sectors, IIT Hyderabad
- 13-18/01/2020 Extended Higgs scenarios in the case of Non-SUSY and SUSY, total 4 lectures given for the Ph.D students at Institute of Physics Bhubaneswar
- 25/12/2019 Seminar on Perspective of Higgs searches at the LHC at Kindai University, Osaka, Japan
- 03/03/2019 Perspective Supersymmetric Higgs sectors at the LHC, HRI, Prayagraj INDIA
- 17/01/2019 Higgs sectors at the LHC, IOP, Bhubaneswar
- 10/12/2018 Perspective Supersymmetric Higgs sectors at the LHC, INFN Lecce, Italy
- 01/11/2018 Distinguishing different Higgs sectors at the LHC, KIAS, Seoul
- 02/08/2018 Standard Model and Higgs boson(s), a seminar given at IISER Kolkata, India
- 09/05/2018 Supersymmetric extended Higgs sectors, University of Pittsburgh, USA
- 06/03/2018 Extended Higgs sectors at the LHC, HRI, Allahabad, India
- 27/06/2018 Higgs and higgsinos in extended supersymmetric Standard Model at the LHC a seminar given at IMSc, Chennai, India
- 15/05/2018 Supersymmetric extended Higgs sectors in at the LHC, a seminar given at University of Pittsburgh, USA
- 07/05/2018 Charged Higgs bosons in supersymmetric extended Higgs sectors at the LHC, a seminar given at Pheno 2018, University of Pittsburgh, USA
- 04/05/2018 Extended Higgs sectors in context of supersymmetry at the LHC, a seminar given at Washington University in St. Louis, USA
- 21/03/2018 Perspective of extended Higgs sectors in beyond Standard Model scenarios, a seminar given at IIT Hyderabad, India
- 09/03/2018 Perspective of extended Higgs sectors in beyond Standard Model scenarios, a seminar given at Sangam@HRI, Allahabad, India
- 28/01/2018 The extended Higgs sectors in supersymmetric scenarios at the LHC, a seminar given at Top@IISERK, IISER Kolkata, India

06-10/11/2017	Charged Higgs bosons in the extended Higgs sectors at the LHC, 7 th KIAS phenomenological workshop, KIAS, Seoul, South Korea
27-30/06/2016	Higgs bosons in extended supersymmetric scenarios at the LHC, Martina Franka, Italy
13-18/06/2016	Charged Higgs bosons in extended supersymmetric scenarios at the LHC, LHCP, Lund University, Sweden
7th March, 2016	Status of extended Higgs sectors in supersymmetric context at the LHC, INFN at University of Roma Tre, Rome, Italy
1st March, 2016	Standard Model and the Higgs boson(s), Indian Institute of Technology, Hyderabad, India
26th February, 2016	Standard Model and the Higgs boson(s), A colloquium at Institute for Mathematical Sciences, Chennai, India
25th February, 2016	Status of extended Higgs sectors in supersymmetric context at the LHC, Institute for Mathematical Sciences, Chennai, India
5th February, 2016	Hidden Higgs boson in extended supersymmetric scenarios at the LHC, Institute for Basics Sciences (CTPU), Daejeon, South Korea
30/01-04/02, 2016	Hidden Higgs boson in extended supersymmetric scenarios, Workshop on Particle physics, String theory and Cosmology, 30/01-04/02, High1 Resort, South Korea
28th September, 2015	Higgs bosons: discovered and hidden, in extended supersymmetric standard models at the LHC at Jawaharlal Nehru University (via Skype), Delhi, India
5th September, 2015	Higgs bosons: discovered and hidden, in extended supersymmetric standard models at the LHC at Summer School and Workshop on the Standard Model and Beyond, Corfu, Greece
22th July, 2015	Higgs bosons: discovered and hidden, in extended supersymmetric standard models at the LHC, SN Bose center for Basic Sciences, Kolkata, India
20th July, 2015	Higgs bosons: discovered and hidden, in extended supersymmetric standard models at the LHC, Institute for Mathematical Sciences, Chennai, India
17th July, 2015	Higgs bosons: discovered and hidden, in extended supersymmetric standard models at the LHC, IITM, Chennai, India
10th July, 2015	Higgs bosons: discovered and hidden, in extended supersymmetric standard models at the LHC, IACS, Kolkata, India
20th March 2015	Higgs physics and supersymmetry at the Large Hadron Collider, IIT Bhubaneswar, Bhubaneswar, India
19th March 2015	Higgs(es) in extended supersymmetric standard model at the LHC, Institute of Physics, Bhubaneswar, India
18th March 2015	Higgs physics and supersymmetry at the Large Hadron Collider, NISER Bhubaneswar, Bhubaneswar, India
28th November, 2014	Charged Higgs(es) in Triplet extended supersymmetric standard model at the LHC, One day workshop at INFN Bari, Bari, Italy
26th October, 2014	Higgs(es) in Triplet extended supersymmetric standard model at the LHC, Particle Physics Day, Jyvaskyula, Finland
22th October, 2014	Higgs(es) in Triplet extended supersymmetric standard model at the LHC, Skype Seminar, Institute of Physics, Bhubaneswar, India

21th October, 2014	Charged Higgs(es) in Triplet extended supersymmetric standard model at the LHC, Particle Physics group Seminar, University of Helsinki, Helsinki, Finland
March, 11-13, 2014	Charged Higgs signature in Triplet Extended Supersymmetric Standard Model at the LHC, Physics Days 2014, Tampere, Finland.
January, 2-7, 2014	Status of Higgses in CP-violating MSSM at the Large Hadron Collider, Spaatind 2014, Skeikampen, Norway.
November, 26, 2013	Status of Triplet extended Supersymmetric Scenarios at the LHC, University of Jyväskylä, Finland.
November, 14, 2013	Status of Triplet extended Supersymmetric Scenarios at the LHC, Karlsruhe Institute of Technology, Karlsruhe, Germany.
November, 11, 2013	Status of Triplet extended Supersymmetric Scenarios at the LHC, DESY, Theory Seminar, Hamburg, Germany.
October, 24, 2013	Two Higgs production in the CP-violating MSSM at the LHC, Physics Day, Helsinki Institute of Physics, Helsinki, Finland
August, 29, 2013	Perspective of Higgs searches in the CP-violating MSSM at the LHC, SUSY 2013, ICTP, Italy.
August, 5, 2013	Higgs Physics beyond Standard Model in the light of LHC results, Indian Institute of Technology Gandhinagar, Gandhinagar, India
August, 2, 2013	Status of Extended Supersymmetric Scenarios at the LHC, Physical Research Institute Ahmedabad, Ahmedabad, India
July, 30, 2013	Status of Extended Supersymmetric Scenarios at the LHC, Indian Institute of Science Bengaluru, Bengaluru, India
February, 21, 2013	Higgs Physics beyond Standard Model in the light of LHC results, IISER Pune, Pune, India
February, 19, 2013	Higgs Physics beyond Standard Model in the light of LHC results, IISER Thiruvananthapuram, Thiruvananthapuram, India
February, 14, 2013	Higgs Physics beyond Standard Model in the light of LHC results, Institute for Mathematical Sciences, Chennai, India
January, 28, 2013	Perspective of Higgs searches at the Large Hadron Collider, IISER Mohali, Chandigarh, India
January, 25, 2013	Limits on MSSM from LHC 2012, part II, University of Helsinki, Helsinki, Finland
January, 18, 2013	Limits on MSSM from LHC 2012, University of Helsinki, Helsinki, Finland
September, 10-14, 2012	Probing Higgs in Type III and Inverse Seesaw at the LHC, 2nd KIAS Phenomenological Workshop, KIAS, Seoul, South Korea (plenary)
August, 12-18, 2012	Probing Higgs in Type III Seesaw at the LHCSUSY 2012, Peking University, Beijing, China
August, 2012	Perspective of Higgs searches at the LHC, IIT Bombay, Mumbai, India
July, 2012	Perspective of Higgs searches at the LHC, IIT Guwahati, Guwahati, India
March, 30, 2012	Mass Measurements at colliders: Lec-4, University of Helsinki, Helsinki, Finland
March, 23, 2012	Mass Measurements at colliders: Lec-3, University of Helsinki, Helsinki, Finland

February, 24, 2012	Mass Measurements at colliders: Lec-2, University of Helsinki, Helsinki, Finland
February, 03, 2012	Mass Measurements at colliders: an Introduction, University of Helsinki, Helsinki, Finland
November, 2011	Probing Higgs in Type III Seesaw at the LHC, KIAS Phenomenological workshop, IACS, Kolkata, India
November, 2011	Probing Higgs in Type III Seesaw at the LHC, KIAS Phenomenological workshop, KIAS, Seoul, Korea
September, 2011	Aspects of Higgs searches in CP-violating MSSM at the Large Hadron Collider, IPMU Seminar, IPMU, Tokyo University, Japan.
September, 2011	Aspects of Higgs searches in CP-violating MSSM at the Large Hadron Collider, IPMU-YITP School and Workshop on Monte Carlo Tools for LHC, YITP, Kyoto University, Japan.
August, 9 - 11, 2011	Aspects of Higgs searches in CP-violating MSSM at the Large Hadron Collider, APCTP 2011 LHC Physics Workshop, Konkuk University, Seoul, Korea.
June, 2011	Aspects of Higgs searches in CP-violating MSSM at the Large Hadron Collider, University of Calcutta, Kolkata.
June, 2011	Aspects of Higgs searches in CP-violating MSSM at the Large Hadron Collider, HIP, University of Helsinki, Finland.
December, 2010	Displaced Higgs production in type III seesaw at the LHC, 21st Busan Particle Physics Workshop, Busan, Korea
November, 2010	Displaced Higgs production in type III seesaw at the LHC, at 'Introduction to physics at the LHC', KIAS, Seoul, South Korea
August, 2010	Displaced Higgs production in type III seesaw at the LHC, TPSC seminar, Delhi University, Delhi, India
August, 2010	Higgs production in CP-violating supersymmetric cascade decays: probing the 'open hole' at the Large Hadron Collider, (parallel session seminar) at SUSY 10, University of Bonn, Bonn, Germany
August, 2010	Displaced Higgs production in type III seesaw at the LHC, APCTP 2011 LHC Physics Workshop, Konkuk University, Seoul, Korea.
July, 2010	Displaced Higgs production in type III seesaw at the LHC, Theory seminar, IISER Kolkata, Kolkata, India
June, 2010	Displaced Higgs production in type III seesaw at the LHC, Theory seminar, Helsinki Institute of Physics, Helsinki, Finland
March, 2010	Aspects of Higgs searches in CP- conserving and CP-violating SUSY scenarios at the LHC and ILC, LCWS 2010, Beijing, China
January, 2010	Higgs searches under Supersymmetric cascades with non-universal gaugino masses, Theory seminar, KIAS, Seoul, South Korea
January, 2010	Status of Higgs boson at the LHC, IIT Bombay, Mumbai, 2010
January, 2010	Higgs searches under Supersymmetric cascades with non-universal gaugino masses, parallel session, WHEPP XI, PRL, Ahmedabad, India

- November, 2009 Higgs searches under Supersymmetric cascades with non-universal gaugino masses, TPSC seminar, University of Delhi, Delhi, India
- November, 2009 Higgs searches under Supersymmetric cascades with non-universal gaugino masses, University of Southampton, Southampton, U.K
- October, 2009 Higgs searches under Supersymmetric cascades with non-universal gaugino masses, University of Manchester, Manchester, U.K
- October, 2009 Higgs searches under Supersymmetric cascades with non-universal gaugino masses, IPPP, Durham, U.K
- October, 2009 Higgs searches under Supersymmetric cascades with non-universal gaugino masses, Helsinki Institute of Physics, Helsinki, Finland
- October, 2009 Higgs searches under Supersymmetric cascades with non-universal gaugino masses, RWTH Aachen University, Aachen, Germany
- October, 2009 Higgs searches under Supersymmetric cascades with non-universal gaugino masses, Bethe seminar, University of Bonn, Bonn, Germany
- February, 2009 Aspects of Higgs searches in scenarios beyond the Standard Model at the Large Hadron Collider, HRI, Allahabad, India
- September 2008 Associated Higgs production in CP-violating supersymmetry: Probing the 'open hole' at the large hadron collider, LHC workshop, HRI, Allahabad, India
- September 2008 Associated Higgs production in CP-violating supersymmetry: Probing the 'open hole' at the large hadron collider, Theory seminar, University of Wisconsin, Madison, USA
- September 2008 Associated Higgs production in CP-violating supersymmetry: Probing the 'open hole' at the large hadron collider, Theory seminar, University of Florida, Gainesville, USA
- September 2008 Associated Higgs production in CP-violating supersymmetry: Probing the 'open hole' at the large hadron collider, Theory seminar, Oklahoma State University, Stillwater, USA
- August 2008 Associated Higgs production in CP-violating supersymmetry: Probing the 'open hole' at the large hadron collider, Theory seminar, SLAC, Menlo Park, USA
- March 2008 Cascade Higgs production at the LHC, HRI, Allahabad
- February 2008 Associated Higgs production in CP-violating supersymmetry: Probing the 'open hole' at the large hadron collider, Higgs hunter's meeting, Viswabharati, Shantiniketan, India
- December 2007 Associated Higgs production in CP-violating supersymmetry: Probing the 'open hole' at the large hadron collider, SINP, Kolkata, India
- December 2007 Associated Higgs production in CP-violating supersymmetry: Probing the 'open hole' at the large hadron collider, IACS, Kolkata, India
- March 2007 CP-Violating-MSSM and signals at the LHC, HRI, Allahabad
- May 2006 Supernova Neutrinos, HRI, Allahabad
- April 2006 Beta function in QCD, HRI, Allahabad
- March 2006 Spin information at LHC, HRI, Allahabad
- November 2005 Neutrinos and their detection, HRI, Allahabad
- November 2005 Solitonic solution in (1+1) dimension, HRI, Allahabad

Poster Presentation

- July, 2014 Higgs bosons: discovered and hidden, in extended supersymmetric standard models at the LHC. Higgs(es) in Triplet extended supersymmetric standard model at the LHC, ICHEP2014, Valencia, Spain
- March, 2013 Perspective of Higgs searches in CPV-MSSM at the LHC, Physics Day conference, Espoo, Finland
- September, 2011 Aspects of Higgs searches in CP-violating MSSM at the Large Hadron Collider, IPMU-YITP School and Workshop on Monte Carlo Tools for LHC, YITP, Kyoto University, Japan.
- September-October, 2010 Displaced Higgs production in type III seesaw at the LHC, Cosmo/Cospa 2010, Tokyo University, Tokyo, Japan